

Magnetic fossils suggest an ancient creature could navigate 97 million years ago

From: ScienceDaily Strange & Offbeat Mined: 2026-10-03 5 min read

Birds, fish and insects can use Earth's magnetic field to help find their way. This ability, called **magnetoreception**, is still one of nature's least understood senses. Researchers now suggest that microscopic fossils may preserve evidence of a magnetic navigation system from 97 million years ago.

The structures, known as **magnetofossils**, were recovered from ancient seafloor sediments in the North Atlantic Ocean. They resemble spearheads, spindles, bullets and needles, and each is no larger than a bacterial cell. Scientists believe they have a biological origin, but the organism that produced them remains unknown.

Researchers from the University of Cambridge and the Helmholtz Zentrum Berlin created the first three-dimensional images of the magnetic structure inside these fossils. Their analysis revealed features that appear particularly suited to detecting both the direction and strength of Earth's magnetic field. Together, those signals could have helped a creature determine its location and direction of travel.

The findings, reported in *Communications Earth & Environment*, support the possibility that the fossils were part of an ancient animal navigation system. Navigation means finding your location and your way from one place to another.

One leading explanation for magnetic sensing involves **magnetite**, a magnetic mineral that may form tiny crystals inside animals. These crystals could align with Earth's field, acting like microscopic compass needles. Some bacteria already use chains of magnetic particles to orient themselves and swim toward their preferred water depth.

The bacterial particles are about 50 to 100 **nanometers** wide. A **nanometer** is one billionth of a meter. The **magnetofossils** in this study are roughly 10 to 20 times larger than the particles bacteria use. Their unusual size had previously led scientists to suggest that they might have served as protective spines.

Computer **simulations** hinted that the structures had more complex magnetic properties. To investigate, the researchers used a new imaging method that maps the arrangement of tiny magnetic fields throughout a particle. Earlier techniques using conventional X rays could not effectively penetrate these large **magnetofossils** to produce such images.

The work used a technique developed by Claire Donnelly at the Max Planck Institute, with measurements performed at the Diamond X-ray facility in Oxford.

The resulting magnetic maps support the navigation explanation, but they do not identify the creature. The study offers a possible window into the **evolution** of magnetic sensing while leaving an important mystery for future research.

Word Treasure

magnetoreception -- The ability to sense Earth's magnetic field.

magnetofossils -- Tiny magnetic structures preserved from ancient life.

magnetite -- A magnetic mineral that may help animals sense magnetic fields.
nanometer -- A unit of length equal to one billionth of a meter.
simulations -- Computer models used to explore how something might work.
evolution -- Changes in living things over many generations.

Background you need

Birds, fish and insects can use Earth's magnetic field to help navigate.

Magnetite crystals could act like microscopic compass needles.

The study's findings were reported in Communications Earth & Environment.

Break it down

Introduces magnetoreception and the 97-million-year-old evidence.

Describes the magnetofossils and their possible navigation function.

Explains magnetite sensing and compares bacterial particles with the fossils.

Describes the imaging challenge, new technique and measurement location.

Ends with the unknown organism and the study's possible contribution to understanding magnetic sensing.

Why it matters

The study may help explain the evolution of magnetic sensing by examining ancient structures that appear suited to navigation.

Test what you remember

1. What is magnetoreception?

- ☐ (A) The ability to sense changes in water temperature.
- ☐ (B) The ability to sense distant sounds underwater.
- ☐ (C) The ability to sense light through thick sediment.
- ☐ (D) The ability to sense Earth's magnetic field.

2. What did the three-dimensional images suggest about the fossils?

- ☐ (A) They could detect magnetic direction and strength.
- ☐ (B) They could detect water movement and pressure.
- ☐ (C) They could detect ocean temperature and depth.
- ☐ (D) They could detect sunlight direction and brightness.

3. What do magnetic particle chains help some bacteria do?

- ☐ (A) Make thicker layers of seafloor sediment.
- ☐ (B) Swim toward their preferred water depth.
- ☐ (C) Grow larger protective spines around themselves.
- ☐ (D) Change the strength of Earth's magnetic field.

4. What purpose had scientists previously suggested for the large magnetofossils?

- ☐ (A) They might have served as protective spines.
- ☐ (B) They might have served as light-sensitive eyes.
- ☐ (C) They might have served as sound-making organs.
- ☐ (D) They might have served as food-storage chambers.

5. Why could earlier techniques not produce the needed magnetic images?

- ☐ (A) Conventional X rays could not effectively penetrate the particles.
- ☐ (B) Researchers could not locate the fossils' original ocean sediments.
- ☐ (C) Computer simulations could not calculate the fossils' exact age.
- ☐ (D) Magnetic minerals could not align with Earth's magnetic field.

6. What contribution did Claire Donnelly make to the work?

- ☐ (A) She discovered the fossils' protective function.
- ☐ (B) She developed the technique used in the study.
- ☐ (C) She identified the organism that made the fossils.
- ☐ (D) She established the preferred depth of bacteria.

Answer key (try the quiz first!): 1: D 2: A 3: B 4: A 5: A 6: B

MY THOUGHTS (at least 20 words):